

Correlation of diffusion in lumbar intervertebral disks with occlusion of lumbar arteries: a study in adult volunteers.

PURPOSE: To evaluate the correlation of the diffusion values in lumbar intervertebral disks with lumbar artery status and the degree of disk degeneration. **MATERIALS AND METHODS:** Sagittal T2-weighted images of the lumbar spine were obtained in 37 asymptomatic volunteers aged 22-68 years. The apparent diffusion coefficient (ADC) of 98 lumbar intervertebral disks was determined, and two-dimensional time-of-flight magnetic resonance angiography was performed on the corresponding 98 lumbar artery pairs (total arteries = 196). The degree of disk degeneration and the status of lumbar arteries were evaluated independently by two radiologists. ADC calculations were performed on the basis of the average signal intensities of the selected region of interest in lumbar disks. The association between ADC values of disks, the disk degeneration, and the status of lumbar arteries of the same level were analyzed with analysis of covariance, and pairwise analysis between groups (Scheffé post hoc multiple comparison) was performed with statistical software. P values less than .01 were considered significant. **RESULTS:** The lumbar arterial status correlated strongly with the diffusion values of intervertebral disks, and the ADC values decreased with higher degrees of arterial narrowing. The correlation between disk degeneration and diffusion was not significant. Eight severely degenerated disks with normal lumbar artery status and diffusion values were found. **CONCLUSION:** Impaired flow in lumbar arteries is significantly associated with decreased diffusion in lumbar disks and may play an important role in disk degeneration.